

Lecture 8: Controllability & Observability

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Disclaimer: These (informal) notes are not intended to replace the textbook or the material contained therein that students are expected to cover and study. If you see any typos, please let me know.

1 Outline

In the previous lecture..

- State-space representation:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \quad x(0) = x_0 \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

- $x(t) \in \mathbb{R}^{n \times 1}$ state vector ($n \geq 2$)
- $u(t) \in \mathbb{R}^{m \times 1}$ input vector
- $y(t) \in \mathbb{R}^{p \times 1}$ output vector
- $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, $D \in \mathbb{R}^{p \times m}$ (usually: $D = 0$)

Today:

- Exponential function
- Observability
- Controllability

Table 1: Today's Ordlista

English	Svenska
state-space (form/representation)	tillstånds (form/beskrivning)
observability matrix	observability matrix
state vector	tillståndsvectorn
observability	observerbarhet
controllability	styrbarhet
identity matrix	enhetsmatrisen
range/image space	värderummet
null space	nollrummet
controllability matrix	styrbarhetsmatrisen
observability matrix	observerbarhetsmatrisen
controllable state	styrbar (tillståndsvektor)
observable state	observerbar (tillståndsvektor)

2 The Matrix Exponential and the Solution of the State Equation

Goal: Solve (i.e., find the solution) of the state-space equation

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t), & x(0) = x_0 \\ y(t) = Cx(t) \end{cases} \quad (1)$$

2.1 Derivation via the Integrating Factor

Consider the state equation $\dot{x}(t) = Ax(t) + Bu(t)$ with $x(0) = x_0$. Multiplying both sides by the integrating factor e^{-At} and recognizing that

$$\frac{d}{dt}(e^{-At}x(t)) = e^{-At}\dot{x} - Ae^{-At}x = e^{-At}(\dot{x} - Ax),$$

the state equation becomes

$$\frac{d}{dt}(e^{-At}x(t)) = e^{-At}Bu(t).$$

Integrating from 0 to t and using $e^{At}e^{-At} = e^{A \cdot 0} = I$ yields:

$$\underbrace{\int_0^t \frac{d}{dt}(e^{-At}x(t)) dt}_{e^{-At}x(t) - x(0)} = \int_0^t e^{-At}Bu(t) dt$$

that is¹,

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau. \quad (2)$$

The first term describes the free evolution from the initial condition x_0 ; the second is the convolution of the matrix exponential with the input, representing the forced response. Note that e^{At} is a *matrix-valued* function of time.

2.2 Computing the Matrix Exponential

The matrix exponential e^{At} satisfies $\frac{d}{dt}e^{At} = Ae^{At}$ with $e^{A \cdot 0} = I$, and may be computed by several equivalent methods².

1. Series expansion.

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \dots$$

2. Inverse Laplace transform. Taking the Laplace transform of $\dot{X} = AX$ with $X(0) = I$ gives $(sI - A)X(s) = I$, hence

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}.$$

¹Exempel 8.10

²Is $e^{At}A = Ae^{At}$? In fact, previously, we had: $\frac{d}{dt}e^{-At} = e^{-At}(-A)$. The answer is yes if we look at the series expansion.

3. Diagonalization. Suppose that A can be diagonalized³, i.e.,

$$A = TDT^{-1}, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

where $D = \text{diag}\{\lambda_1, \dots, \lambda_n\}$ collects the eigenvalues of A , then

$$e^{At} = Te^{Dt}T^{-1}, \quad e^{Dt} = \text{diag}\{e^{\lambda_1 t}, \dots, e^{\lambda_n t}\}.$$

A fundamental stability consequence follows immediately: $e^{At} \rightarrow 0$ as $t \rightarrow \infty$ if and only if all eigenvalues of A lie in the open left-half plane (LHP), consistent with the stability theorems of F7.

4. Cayley-Hamilton's theorem

Theorem 2.1 (Cayley–Hamilton). *Every square matrix satisfies its own characteristic equation. If*

$$p(\lambda) = \det(\lambda I - A) = \lambda^n + a_1\lambda^{n-1} + \dots + a_{n-1}\lambda + a_n,$$

then

$$A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = 0.$$

An immediate corollary is that, when A^{-1} exists (check: $a_n = \det(-A) \neq 0$), then it is given by:

$$A^{-1} = \frac{-1}{a_n}(A^{n-1} + a_1A^{n-2} + \dots + a_{n-1}I).$$

Moreover, Cayley–Hamilton implies that every power A^p with $p \geq n$ can be written as a linear combination of $I, A, \dots, A^{n-1}$:

$$\begin{aligned} A^n &= -a_1A^{n-1} - \dots - a_{n-1}A - a_nI \\ A^{n+1} &= -a_1A^n - \dots - a_{n-1}A^2 - a_nA = b_1A^{n-1} + \dots + b_{n-1}A + b_nI \\ &\vdots \quad \text{Continue for } p \geq n \\ A^p &= b_1^{(p)}A^{n-1} + \dots + b_{n-1}^{(p)}A + b_n^{(p)}I \end{aligned}$$

Substituting this reduction into the series for e^{At} gives:

$$e^{At} = I + At + \dots = f_0(t)I + f_1(t)A + \dots + f_{n-1}(t)A^{n-1}, \quad (3)$$

where $\{f_k(t)\}_{k=0}^{n-1}$ are scalar, linearly independent functions of time.

Example 0: Double integrator

Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Since $A^2 = 0$, the series terminates and $e^{At} = I + At$. Equivalently, via the inverse Laplace method:

$$(sI - A)^{-1} = \frac{1}{s^2} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}, \quad e^{At} = \mathcal{L}^{-1} \left\{ \begin{bmatrix} 1/s & 1/s^2 \\ 0 & 1/s \end{bmatrix} \right\} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

³Note that not all the matrices can be diagonalized: What are the requirements for A to be diagonalizable? Go back to the linear algebra course..

3 Controllability and Observability

Before giving formal definitions, it is instructive to examine a concrete system in which the issues of controllability and observability arise naturally.

Example 1: Exempel 8.13

Consider the system

$$\begin{aligned}\dot{x} &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} u, \\ y &= [2 \quad 1 \quad 0] x.\end{aligned}$$

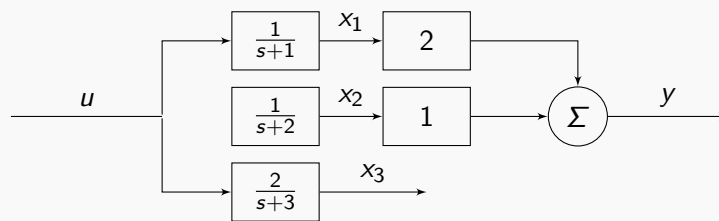
The system transfer function is given by

$$G(s) = C(sI - A)^{-1}B = \frac{2(s+2)(s+3)}{(s+1)(s+2)(s+3)} = \frac{2}{s+1}$$

Do we have the right number of states? Although the state-space model has $n = 3$ states, the transfer function has order one. This discrepancy signals that some states are either not driven by the input or not visible in the output. Examining the individual state equations:

- $\dot{x}_2 = -2x_2 + 0u$: the state x_2 is entirely decoupled from the input. It evolves autonomously as $x_2(t) = e^{-2t}x_2(0)$, independently of any control action. The state x_2 is therefore *not controllable*.
- The coefficient of x_3 in C is zero, so x_3 contributes nothing to/is not visible from the output. The state x_3 is therefore *not observable*.

The pole-zero cancellations in $G(s)$ mask these structural deficiencies, which would be invisible to a purely transfer-function-based analysis. Compare with the block diagram of the system:



Observability:

Can I uniquely determine the entire state vector $x(t)$ of a system from the outputs $y(t)$ over some time interval? Do the sensors (measured outputs) give enough information to fully capture what is happening inside the system?

Are there parts of the system's state that are unobservable, evolving internally but never showing up in the measured outputs?

Is it possible to design an observer that reconstructs the state from outputs?

Controllability:

Starting from some initial condition, can I drive the system to any desired state $x(t_f)$ within finite time using suitable inputs?

Are there internal states that evolve on their own and cannot be influenced by control inputs?

Is it possible to design a controller that stabilizes or regulates the system to a desired behavior?

3.1 Controllability

Definition 3.1 (Controllability, Definition 8.3). A state $x^* \in \mathbb{R}^n$ is *controllable* if there exists a finite time $T > 0$ and an input $u : [0, T] \rightarrow \mathbb{R}^m$ such that $x(0) = 0$ and $x(T) = x^*$. A system is *controllable* if every state $x^* \in \mathbb{R}^n$ is controllable.

3.1.1 Algebraic Test for Controllability

With $x(0) = 0$, the solution (2) gives $x(t) = \int_0^t e^{A(t-\tau)} B u(\tau) d\tau$. Substituting the Cayley–Hamilton representation (3), i.e., $e^{At} = f_0(t)I + \dots + f_{n-1}(t)A^{n-1}$, we obtain

$$x(t) = \gamma_0(t)B + \gamma_1(t)AB + \dots + \gamma_{n-1}(t)A^{n-1}B,$$

where $\gamma_k(t) = \int_0^t f_k(t-\tau)u(\tau) d\tau$ are scalar quantities determined by the input. Every controllable state is therefore a linear combination of the columns of $B, AB, \dots, A^{n-1}B$, i.e., it lies in the range (column) space of the **controllability matrix**

$$S = [B \ AB \ A^2B \ \dots \ A^{n-1}B] \in \mathbb{R}^{n \times nm}. \quad (4)$$

Theorem 3.2 (Controllability criterion, Resultat 8.9). *The system (A, B) is controllable if and only if the controllability matrix S in (4) has full rank n (if S is square: $\det S \neq 0$).*

Example 2: Double integrator

For $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$AB = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad S = [B \ AB] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \det(S) = -1 \neq 0.$$

The system is therefore **controllable**.

3.2 Observability

Definition 3.3 (Observability, Definition 8.4). A state $x^* \in \mathbb{R}^n$ is *unobservable* if, when $x(0) = x^*$ and $u(t) \equiv 0$ (identically zero) for all $t \geq 0$, the output satisfies $y(t) \equiv 0$ for all $t \geq 0$. A system is *observable* if the only unobservable state is $x^* = 0$.

Intuitively, a system is observable when the measured outputs carry sufficient information to uniquely reconstruct the complete internal state vector.

3.2.1 Algebraic Test for Observability

With $u(t) \equiv 0$ and $x(0) = x^*$, the output is $y(t) = Ce^{At}x^*$. Differentiating repeatedly and evaluating at $t = 0$:

$$\begin{aligned} y(0) &= Cx^* = 0, \\ \dot{y}(0) &= CAx^* = 0, \\ &\vdots \\ y^{(n-1)}(0) &= CA^{n-1}x^* = 0. \end{aligned}$$

By Cayley–Hamilton, no new information is contained in derivatives of order n or higher. The unobservable states x^* therefore lie in the null space of the **observability matrix**

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{np \times n}. \quad (5)$$

Theorem 3.4 (Observability criterion, Resultat 8.10). *The system (C, A) is observable if and only if the observability matrix $\mathcal{O}$ in (5) has full rank n (if $\mathcal{O}$ is square: $\det(\mathcal{O}) \neq 0$).*

Example 3: A non-observable system

For $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ and $C = [1 \ 0]$:

$$CA = [0 \ 0], \quad \mathcal{O} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \det(\mathcal{O}) = 0.$$

The system is therefore **not observable**.

Example 4: Observability of Example 2, cont'd

For $A = \text{diag}(-1, -2, -3)$ and $C = [2 \ 1 \ 0]$:

$$CA = [-2 \ -2 \ 0], \quad CA^2 = [2 \ 4 \ 0], \quad \mathcal{O} = \begin{bmatrix} 2 & 1 & 0 \\ -2 & -2 & 0 \\ 2 & 4 & 0 \end{bmatrix}.$$

The third column of $\mathcal{O}$ is identically zero, so $\det(\mathcal{O}) = 0$ and the system is not observable. The null space of $\mathcal{O}$ is spanned by $e_3 = [0 \ 0 \ 1]^T$, confirming that any initial condition $x^* = [0 \ 0 \ x_3]^T$ produces $y(t) \equiv 0$: the state x_3 is the unobservable mode, consistent with the block-diagram analysis of Example 2.

4 Minimal Realization

The previous examples show that a given transfer function $G(s)$ may be realized by state-space models of different orders. This raises the natural question of the smallest possible realization.

Definition 4.1 (Minimal realization). A state-space representation (A, B, C, D) is a *minimal realization* of a transfer function $G(s)$ if no state-space model with fewer states has the same transfer function.

Theorem 4.2 (Resultat 8.12). *A state-space representation (A, B, C, D) is a minimal realization of an input-output relationship (i.e., its transfer function) if and only if it is both controllable and observable.*

This theorem provides a clean interpretation of the pole-zero cancellations encountered in transfer-function algebra. When a transfer function appears to have fewer poles than the associated state-space model, the “missing” poles correspond either to *uncontrollable modes* (states decoupled from the input) or to *unobservable modes* (states invisible to the output). In Example 2, the reduction from third to first order arises precisely from the presence of one uncontrollable state (x_2) and one unobservable state (x_3), each contributing a pole-zero cancellation in $G(s)$.

Remark 4.3 (Duality). Controllability and observability are dual properties in a precise algebraic sense: (A, B) is controllable if and only if $(A^\top, B^\top)$ is observable. This duality principle allows algorithms and theoretical results for one property to be translated directly into those for the other, and is the basis for the parallel structure of observer design (dual to controller design) in subsequent lectures.